

Thesis

- Lehrstuhl für Reliability Engineering
- Prof. Dr. Matthias G. R. Faes

Physics-Informed Extreme Learning Machines for Uncertainty
Quantification in Computational Mechanics

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Studiengang: Maschinenbau

1. Motivation & Problem

2. Extreme Learning Machine (ELM)

- Key idea
- Advantages & limitations

3. Physics-Informed ELM (PIELM)

- Concept and formulation
- Key advantages
- Extension to uncertainty quantification

4. Numerical Examples

- **Cantilever Beam**
 - Deterministic results
 - Parametric studies
 - UQ results
- **String Equation**
 - Deterministic results
 - UQ results

5. Conclusions & Future Work

Motivation

The Challenge



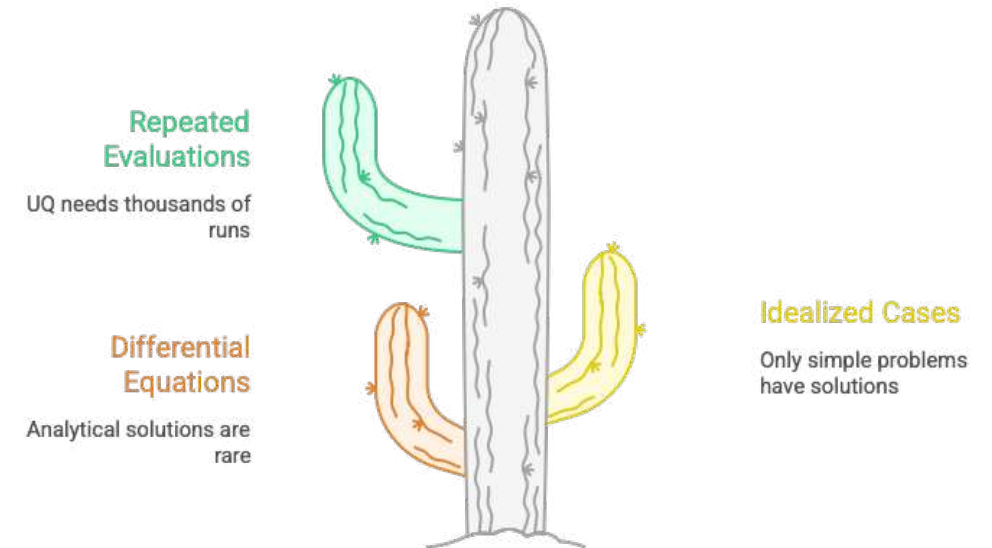
- Engineering systems → governed by **differential equations**
 - Analytical solutions are **rare** (only ideal cases)
 - UQ requires **thousands** of model evaluations
- **High computational cost**

The Core Problem



Traditional numerical solvers (**FEM**, **FDM**)
are **too slow** for repeated evaluations in UQ

Slow Solvers Hinder Uncertainty Quantification



Made with Napkin

Research Gap

- Existing surrogate methods (PCE, Kriging, ANN) suffer from **scalability** and **computational limitations**.
- Physics-informed approaches (PINNs) improve physical consistency but require **expensive iterative optimization**.
- **PIELM** provides **fast** and **physically consistent** solutions
→ but mainly applied to **deterministic problems**.



Novelty of This Work

- First extension of **PIELM** to **uncertainty quantification**.
- Uses augmented input space $(\Omega \times \Theta)$.
- Enables a **single surrogate model** to cover the full uncertainty domain.

Problem Statement

Can PIELM be extended for efficient uncertainty quantification?

Why ELM?



Need for **computationally efficient** surrogate models



Avoid **expensive iterative training**
of neural networks



Suitable for **fast approximation** tasks

What is ELM?



Single hidden layer
neural network

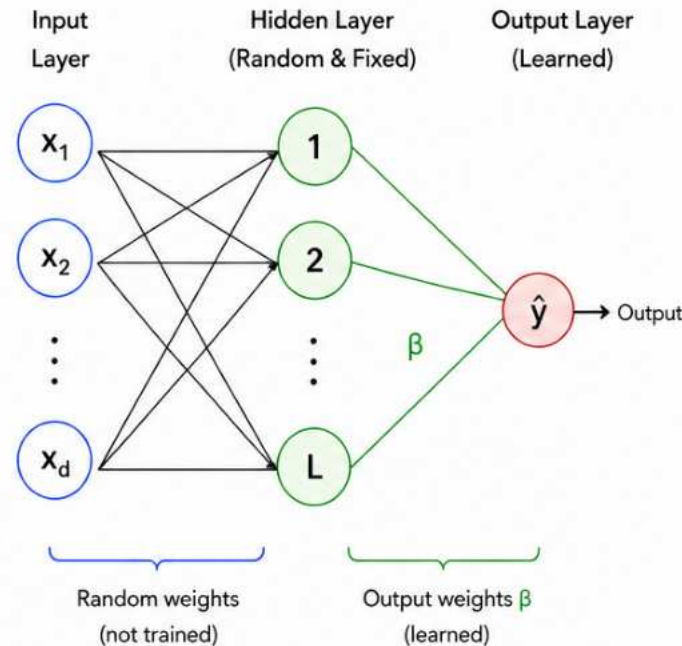


Hidden layer parameters:

- ✓ Randomly assigned
- ✓ Fixed (not trained)



Only output weights
are learned



1. Hidden layer output

$$H = g(WX + b)$$

- $X \in \mathbb{R}^{d \times N}$: input data (N samples)
- $W \in \mathbb{R}^{L \times d}$: input weights (random)
- $b \in \mathbb{R}^L$: biases (random)
- $g(\cdot)$: activation function
- $H \in \mathbb{R}^{L \times N}$: hidden layer output matrix

2. Output layer model

$$H\beta = Y$$

- $\beta \in \mathbb{R}^{L \times m}$: output weights (unknown)
- $Y \in \mathbb{R}^{N \times m}$: target outputs
- m : output dimension

3. Solve for output weights

$$\beta = H^\dagger Y$$

where $H^\dagger$ is the Moore–Penrose pseudoinverse.

(With regularization)

$$\beta = (H^T H + \lambda I)^{-1} H^T Y$$

- $\lambda > 0$: regularization parameter
- I : identity matrix ($L \times L$)

Why ELM?

- ✓ No iterative training (no backpropagation)
- ✓ Very fast training (single linear solve)
- ✓ Good generalization with proper regularization
- ✓ Simple and efficient

ELM: Advantages and Limitations



Advantages



Extremely fast training

Training reduces to solving a linear system
– no iterative optimization.



Deterministic solution

Closed-form solution for output weights
ensures repeatability.



Simple implementation

Easy to implement with low
computational complexity.



Limitations



Does not enforce physical laws

ELM is purely data-driven and does not
incorporate governing equations or
boundary conditions.



May produce non-physical solutions

Predictions can violate physical laws and
boundary conditions, leading to
unreliable results.



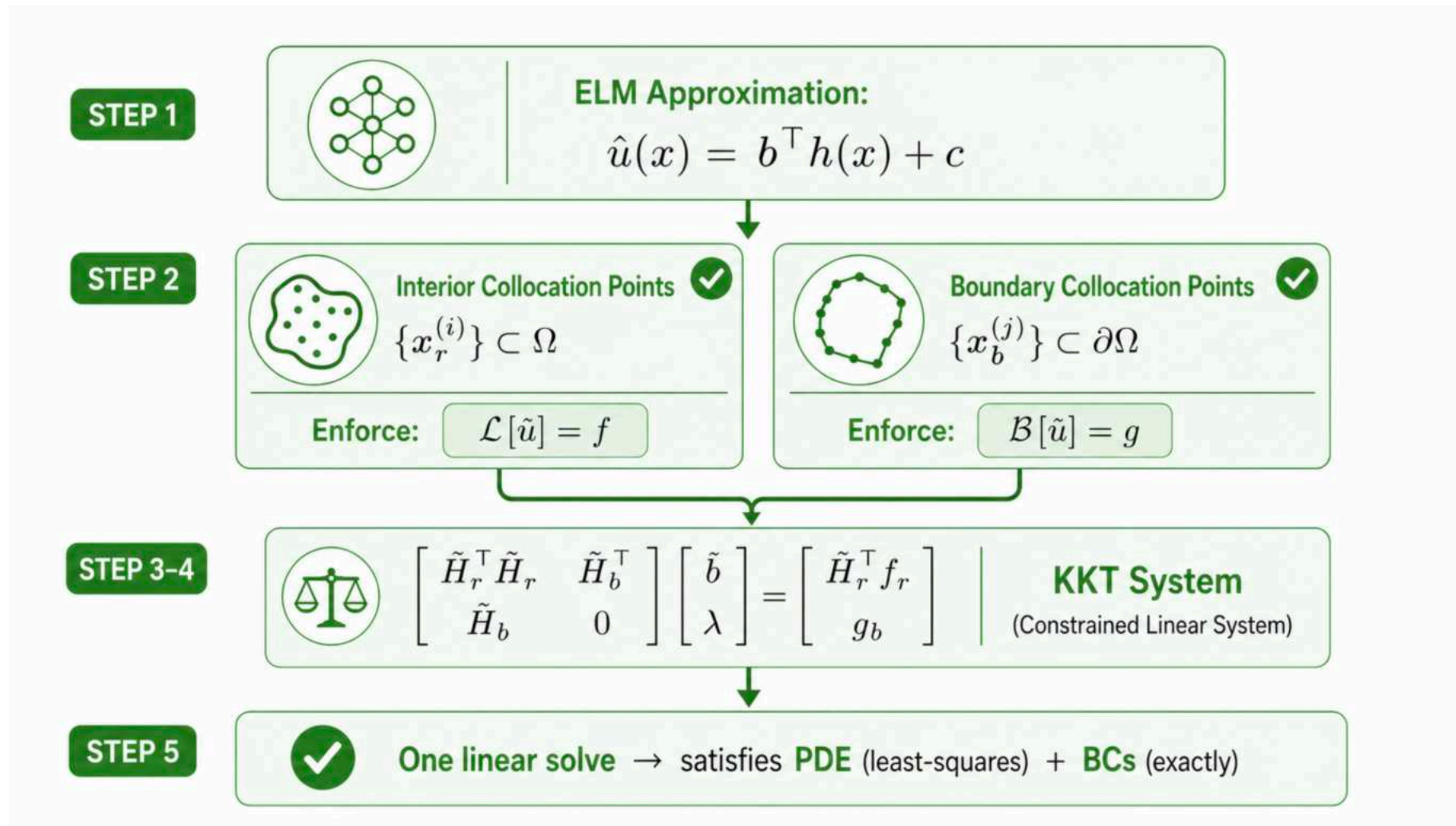
Key Takeaway

ELM is extremely efficient but lacks physical consistency.
This motivates the need for **Physics-Informed ELM (PIELM)**.

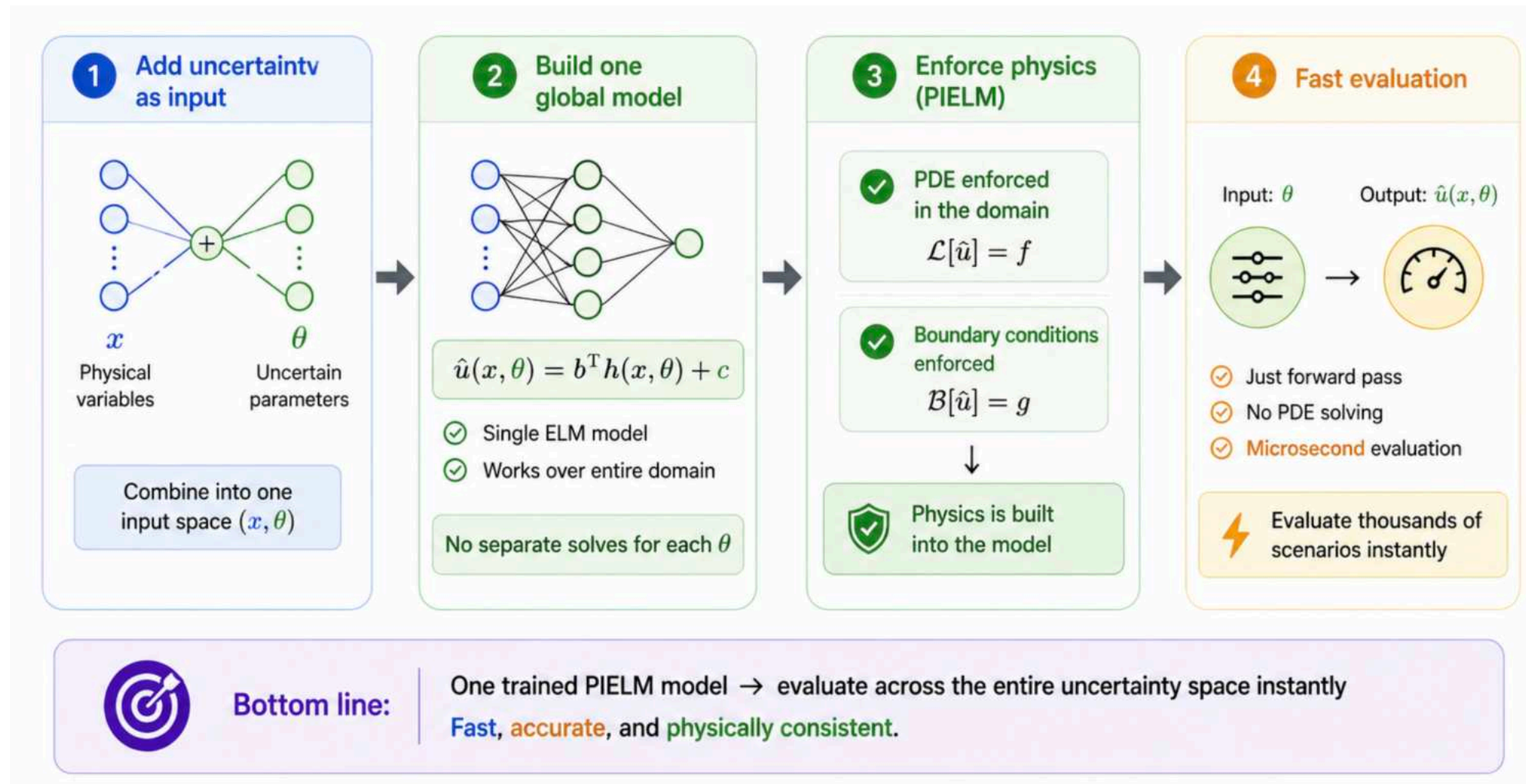
Physics-Informed Extreme Learning Machine (PIELM)



How does PIELM Work?



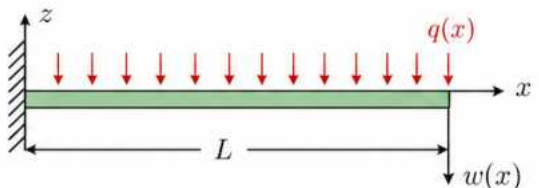
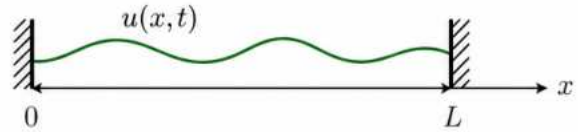
PIELM for Uncertainty Quantification



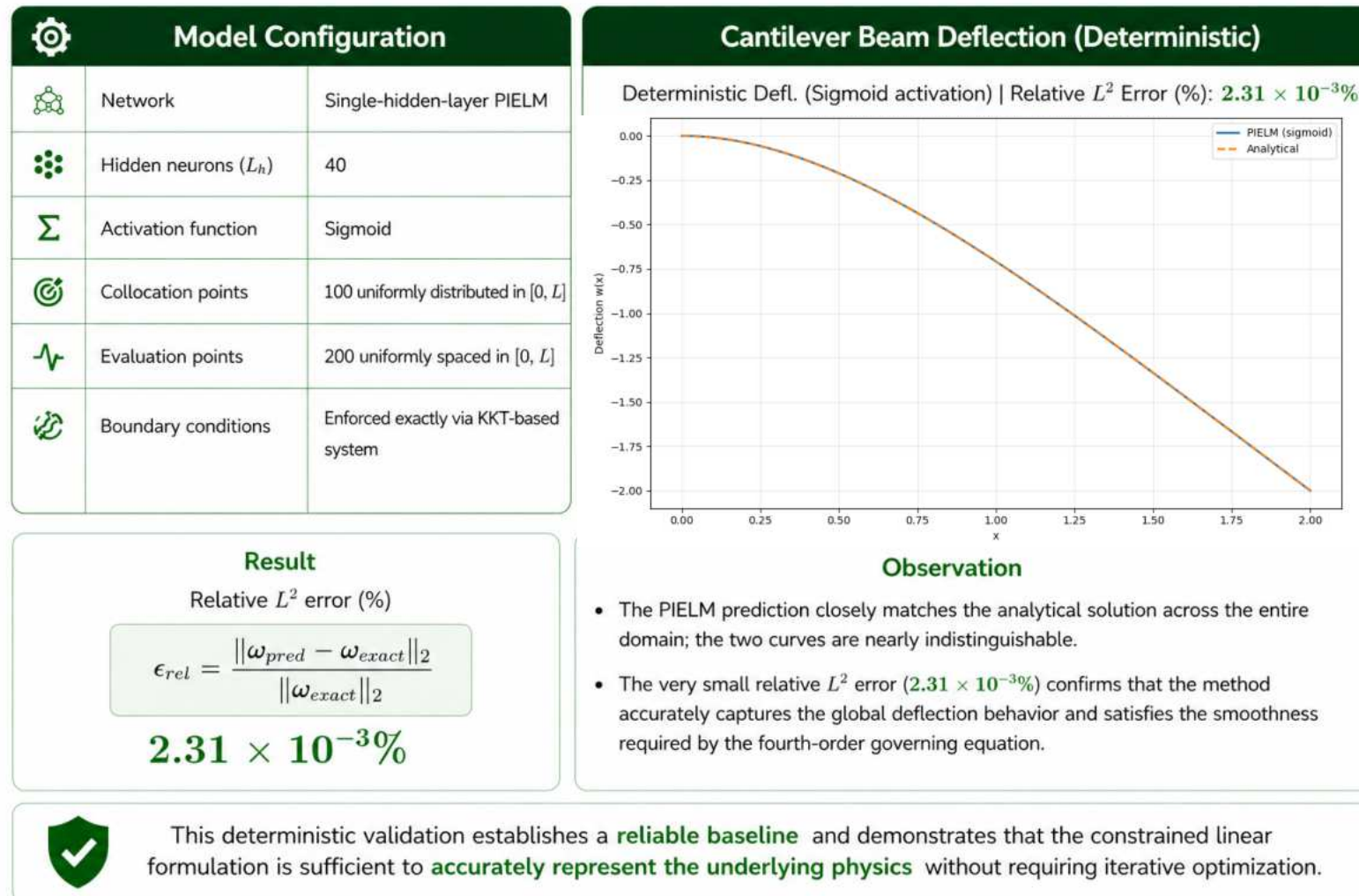
Examples



To verify the proposed PIELM framework, two examples were taken from mechanics. These cover both static and dynamic behaviors.

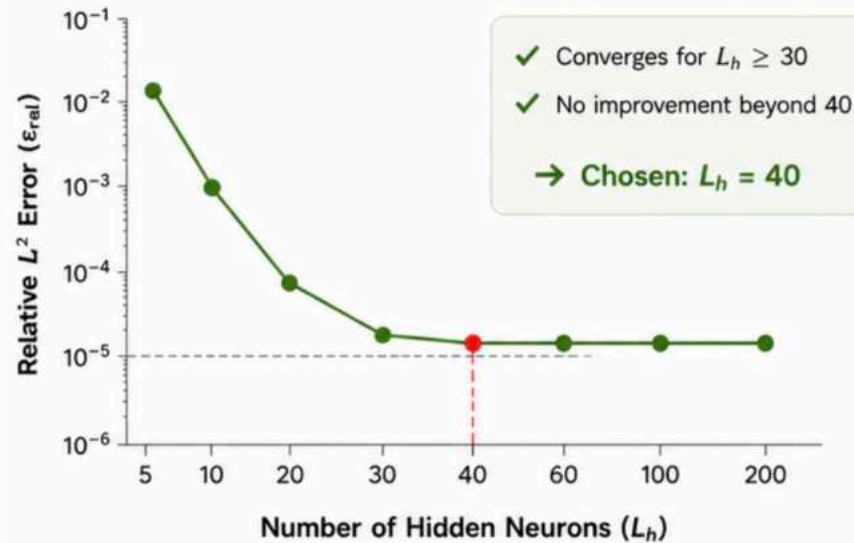
	1 Cantilever Beam (Static)	2 Vibrating String (Dynamic)
 Equation	 4th-order ODE (Static) $EI \frac{d^4 w(x)}{dx^4} = q(x), \quad 0 < x < L$	 Wave equation (dynamic) $\frac{\partial^2 u(x,t)}{\partial t^2} = c^2 \frac{\partial^2 u(x,t)}{\partial x^2}, \quad 0 < x < L, \quad t > 0$
 Uncertainty	$E, I \sim U[0,9,1.1]$ (Young's modulus and moment of inertia)	$c \sim U[0,9,1.1]$ (Wave speed)
 Why?	 Structural mechanics baseline Classic static problem for benchmarking accuracy and efficiency.	 Tests time-dependent behavior Introduces dynamics to validate performance on evolution in time.
 Together, these two problems validate PIELM across static and dynamic regimes under parametric uncertainty.		

Deterministic Result

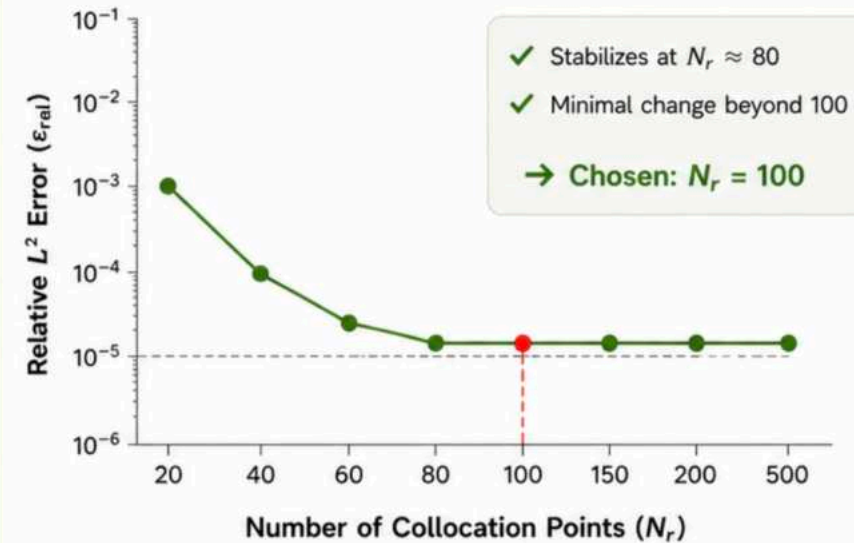


Hyperparameter Convergence

1. Neuron Convergence ($N_r = 100$)



2. Collocation Point Convergence ($L_h = 40$)



Key Takeaway

Parameters are selected in the **converged regime** to achieve high accuracy with **efficient cost**.



Stability Check (Random Initialization)

- 30 runs with different random seeds
- Mean relative L^2 error $\approx 2.3 \times 10^{-5}$
- All runs $< 3.2 \times 10^{-5}$
- ✓ **Model is stable and reliable.**

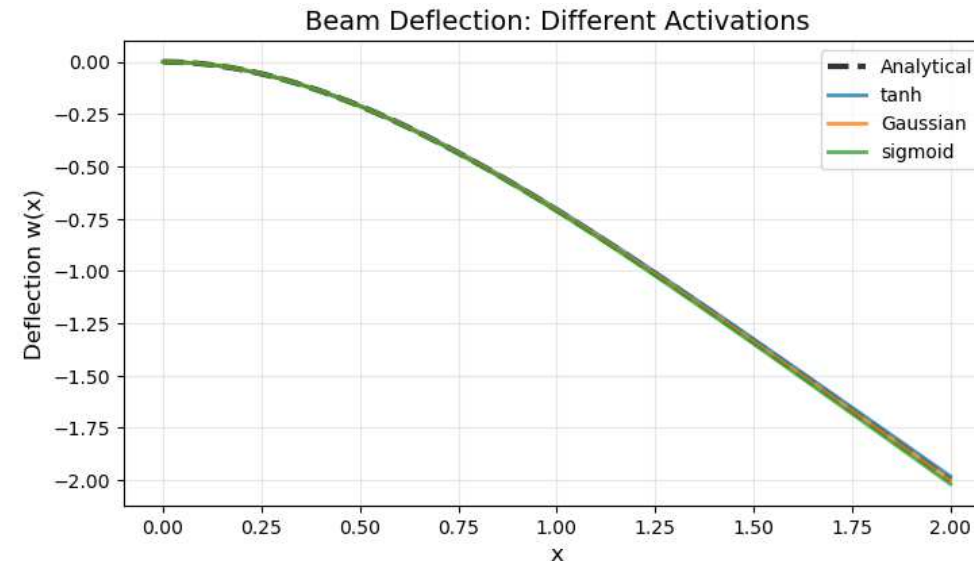
Activation Function Study

Problem Setup

To assess the influence of activation functions on the PIELM approximation, three commonly used nonlinear activation functions are considered:

- sigmoid
- hyperbolic tangent (*tanh*)
- Gaussian

All other parameters and settings are kept identical to those used in the deterministic study, allowing a direct comparison of the effect of the activation function.



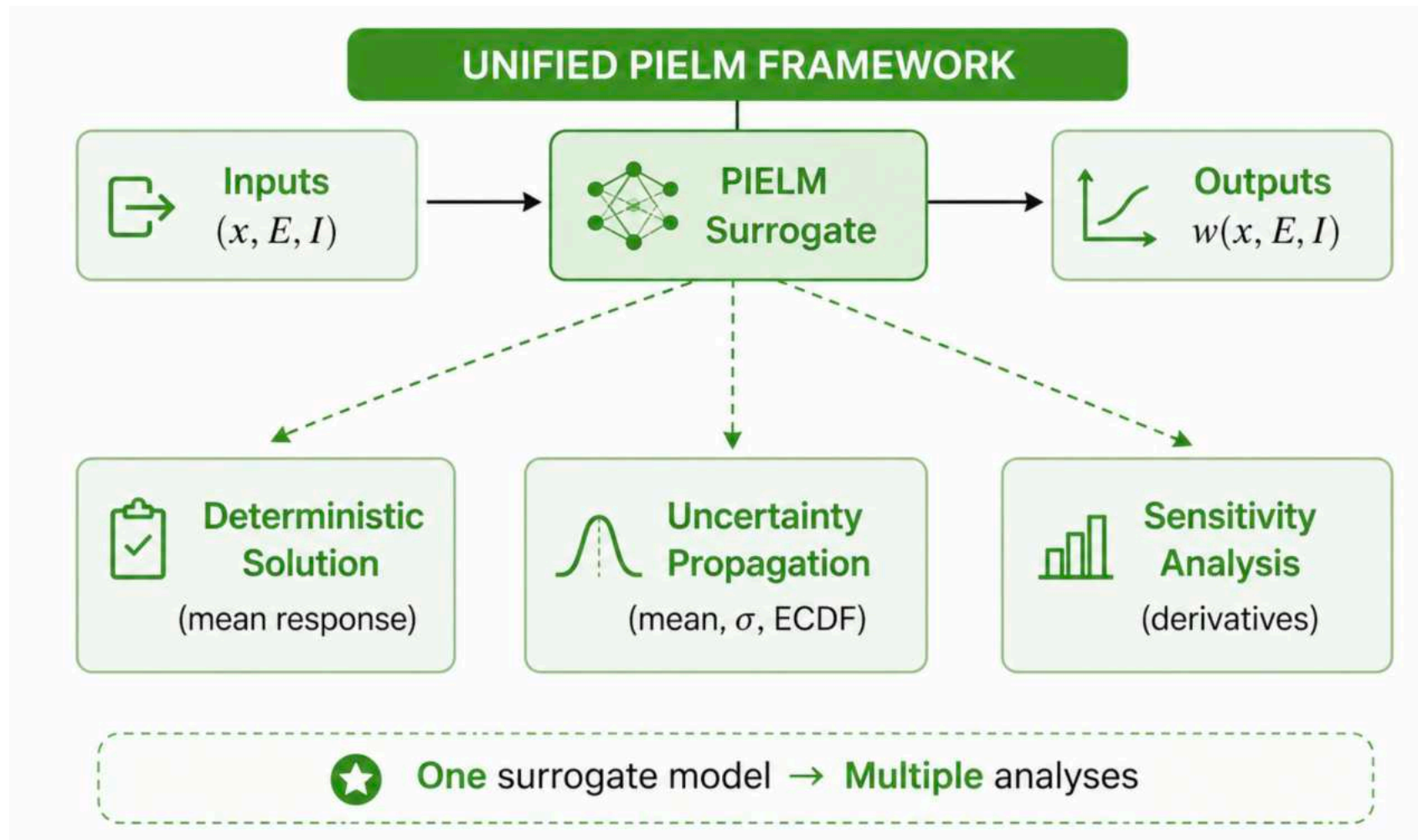
Results

Activation	Relative L^2 Error (%)	Time (s)
tanh	$2.82 \times 10^{-2} \%$	0.1188
sigmoid	$3.46 \times 10^{-3} \%$	0.0167
gaussian	$1.19 \times 10^{-3} \%$	0.0270

Observations

- All three activation functions closely match the analytical solution, as the predicted and analytical curves are nearly indistinguishable.
- The **Gaussian activation** performs best due to its **smooth and infinitely differentiable** nature, which leads to more accurate higher-order derivatives and more stable residual representation for the fourth-order beam equation.
- **Sigmoid** is the fastest, while **Gaussian** has a slightly higher computational cost due to exponential evaluations.
- Overall, the **Gaussian activation** offers the **best balance between accuracy, smoothness, and numerical robustness**, and is adopted for subsequent studies.

PIELM for Uncertainty Quantification



Sampling Strategies



Sampling determines where and how many points we use in the augmented space

$$Z = \Omega \times \Theta$$

1 Accuracy



Captures the variation in the solution across physical and parameter spaces.

2 Stability



Well-distributed points lead to better conditioning and more robust solutions.

3 Efficiency



Achieves good accuracy with fewer points and lower computational cost.



In short: Right points, in the right places, in the right number → **accurate results** with minimal cost.

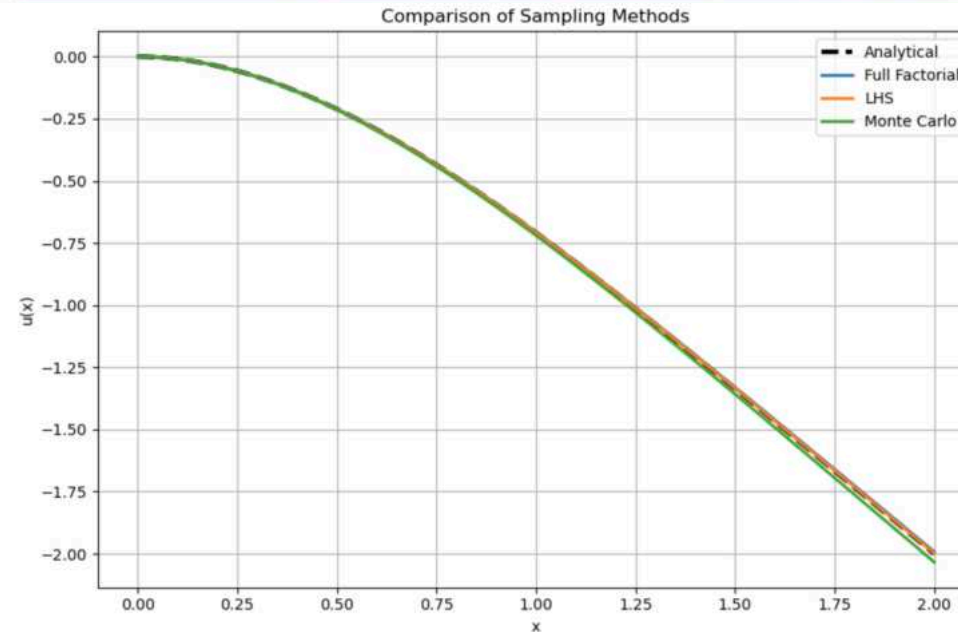
Sampling Strategies: Comparison



To efficiently explore the augmented space (x, E, I) and mitigate the curse of dimensionality, we investigate **three sampling strategies** for collocation point generation.

1. Problem Setup

- **Beam deflection:** $w = w(x, E, I)$
- **Uncertain parameters:**
 $E \sim \mathcal{U}[0.9, 1.1], I \sim \mathcal{U}[0.9, 1.1]$
- **Augmented domain:**
 $(x, E, I) \in [0, L] \times [0.9, 1.1]^2$
- **PIELM model:** 150 hidden neurons
- **Activation function:** Gaussian
- **Interior points:** $N_r = 1000$
- **Boundary points (at $x = L$):** 100
- **Boundary conditions:** Imposed as hard constraints (KKT)



4. Discussion & Key Takeaways



Full Factorial Sampling

- Structured, exhaustive coverage
- Accurate in low dimensions
- Computational cost grows exponentially



Monte Carlo Sampling

- Easy to implement
- Avoids exponential growth
- May cause clustering and local gaps $\rightarrow$ affects conditioning



Latin Hypercube Sampling (LHS)

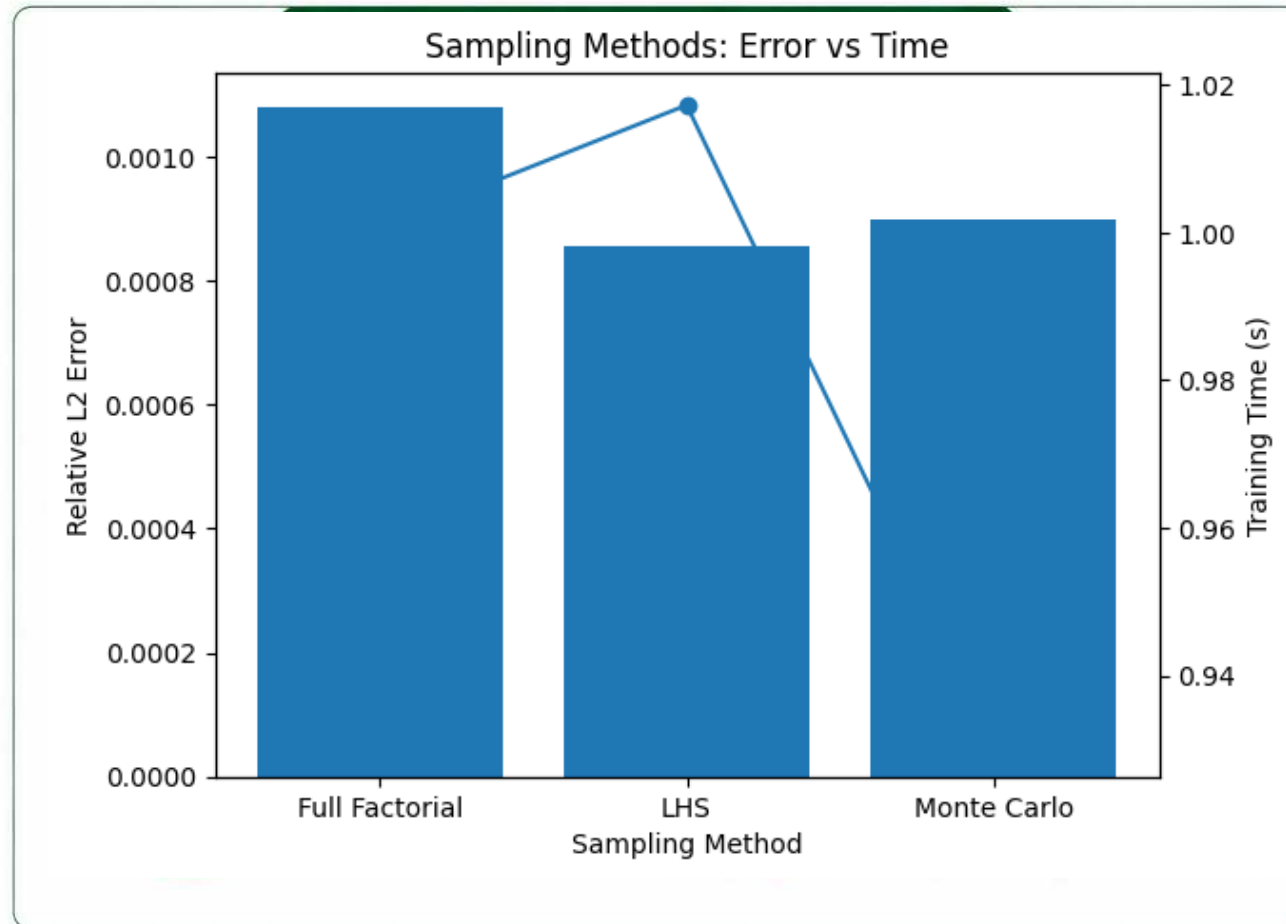
- Uniform marginal coverage
- Better space-filling
- More stable and reliable than Monte Carlo



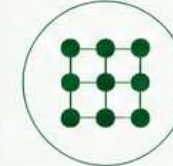
Overall Conclusion

All methods achieve similar accuracy, but **LHS** offers the best balance between coverage, scalability, and robustness.

Sampling Strategies: Comparison

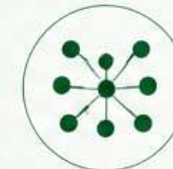


Sampling Strategy Used



LHS for Collocation

- Better domain coverage

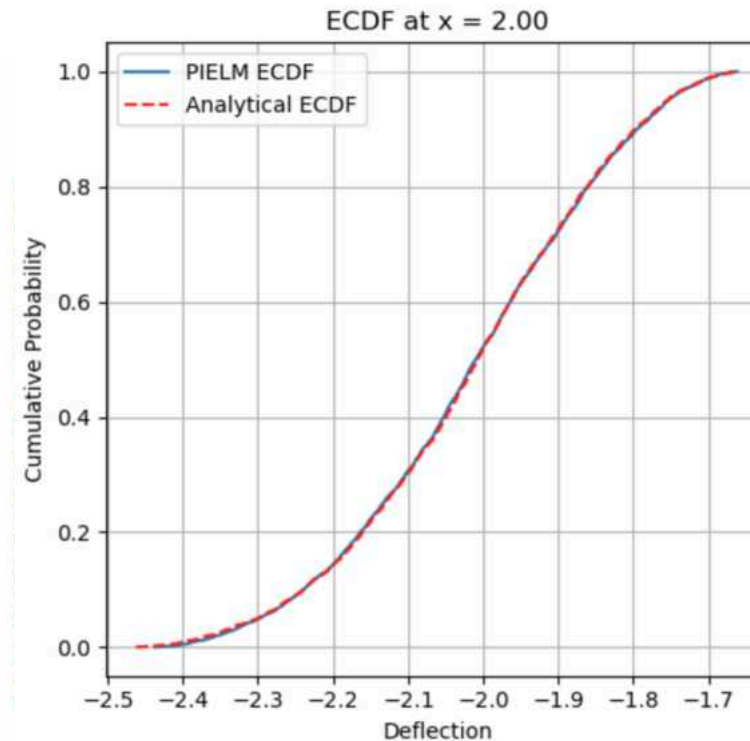
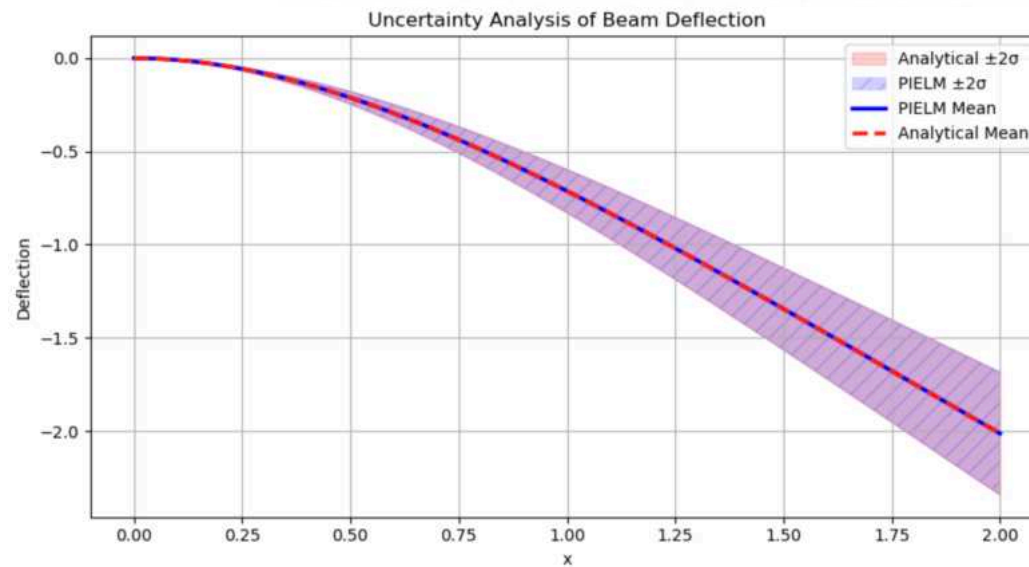


Monte Carlo for UQ

- Statistical reliability
- Robustness
- Ability to capture the **full uncertainty distribution**

Results

1. Mean Deflection with $\pm 2\sigma$ Band



- PIELM matches analytical solution closely
- Max error in mean: 4.2×10^{-6}
- Max error in standard deviation: 1.1×10^{-6}
- Uncertainty increases along the beam length
- PIELM reproduces full probability distribution
- Max ECDF difference < 0.005
- Within Monte Carlo sampling error ($\approx \frac{1}{\sqrt{5000}} \approx 0.014$)
- Captures tail behavior & failure probabilities

Results

Sensitivity Analysis at Midpoint ($x = L/2$)

Quantity	PIELM	Analytical	Rel. Error
$\partial w / \partial E$	-6.869×10^{-1}	-6.875×10^{-1}	0.09%
$\partial w / \partial I$	-7.711×10^{-1}	-7.708×10^{-1}	0.04%
$S_E = \frac{E}{w} \frac{\partial w}{\partial E}$	-0.967	-0.968	0.10%
$S_I = \frac{I}{w} \frac{\partial w}{\partial I}$	-1.085	-1.084	0.09%



- Deflection decreases with $\uparrow E$ and $\uparrow I$ (negative sensitivity)



- Geometry (I) slightly more influential than material (E)



- Relative error $< 0.1\%$ $\rightarrow$ high accuracy



PIELM accurately captures parameter sensitivity and physical trends.

PIELM String Equation Deterministic

PROBLEM SETUP

- Vibrating string problem
- Displacement approximation

$$u = u(x, t, c)$$

where x (space), t (time), c (wave speed) are input variables.

- Deterministic case

$$c = 1.0$$

- Neural network configuration

- Single hidden layer
- $L_{\text{neurons}} = 150$ hidden neurons
- Gaussian activation function
- Random weights and biases (fixed random seed)

- Collocation points (Latin Hypercube Sampling)

- Interior (PDE): $N_r = 6000$ points
- Initial condition: $N_{ic} = 2000$ points

- Boundary conditions

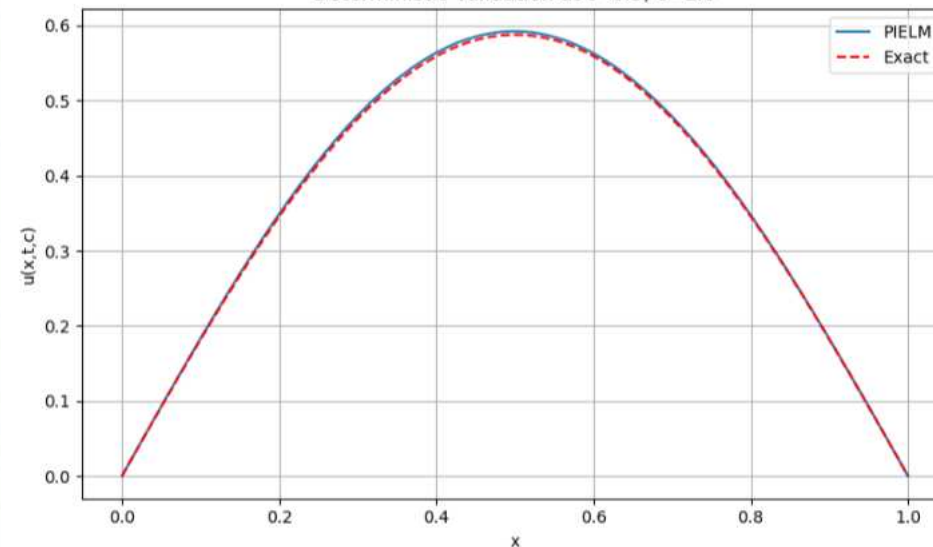
Satisfied exactly by constructing the approximation as

$$\hat{u}(x, t, c) = \sin\left(\frac{\pi x}{L}\right) \hat{u}(x, t, c)$$

where $L = 1.0$ is the string length. This ensures the Dirichlet conditions are identically satisfied for all t and c .

RESULTS

Comparison at $t = 0.3$ and $c = 1.0$



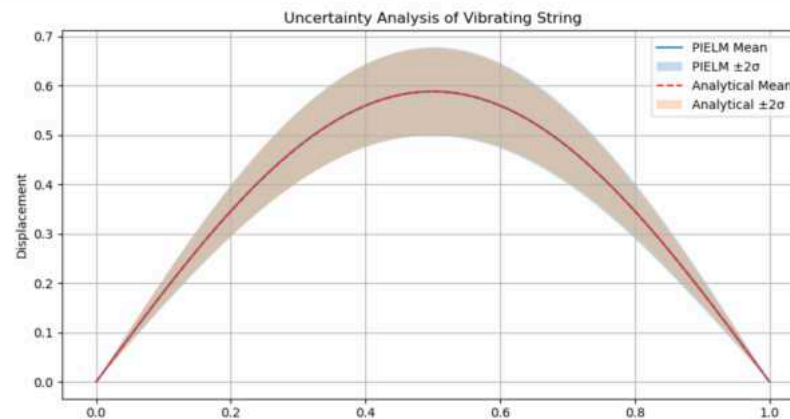
High Accuracy

- Numerical and analytical solutions are nearly indistinguishable.
- Relative L_2 error over $x \in [0, L]$ at $t = 0.3$:

$$\epsilon_{rel} = 1.87 \times 10^{-4}$$

Result

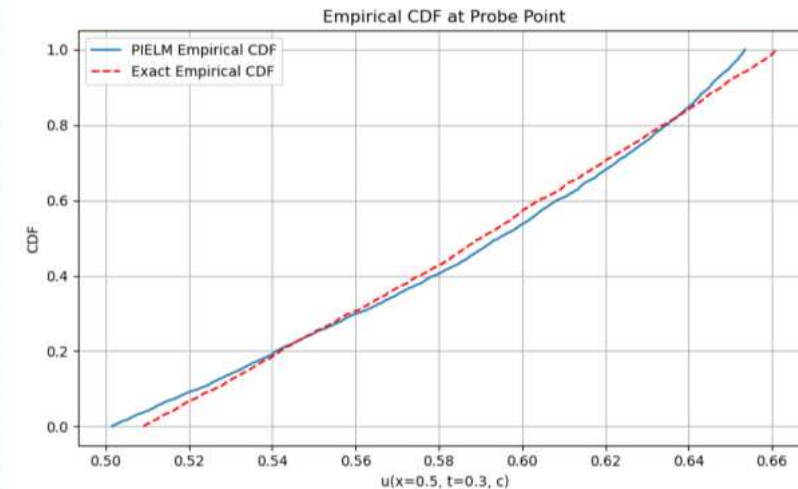
Mean Displacement and Uncertainty Band at $t = 0.3$



Mean Displacement & Uncertainty Band

- PIELM mean closely matches analytical mean.
- Uncertainty band closely agrees with analytical reference.
- Largest variability occurs near the midspan.
- Uncertainty vanishes at fixed boundaries due to Dirichlet conditions.

Empirical CDF at Probe Point ($x = 0.5, t = 0.3$)



Empirical CDF at Probe Point

- PIELM ECDF closely matches analytical distribution.
- Accurate prediction across the entire probability range.
- Captures both central tendency and tail behavior.
- Small deviations due to Monte Carlo sampling variability.
- Confirms reliability of PIELM for probabilistic response modeling.

Result

Sensitivity Analysis at $x = 0.5$, $t = 0.3$, $c = 1.0$

Quantity	PIELM	Analytical	Relative Error
$u(x, t, c)$	5.9210×10^{-1}	5.9210×10^{-1}	$< 0.01\%$
$\frac{\partial u}{\partial c}$	-7.9067×10^{-1}	-7.6248×10^{-1}	3.70%
S_c	-1.3354	-1.2972	2.95%

Key Insights



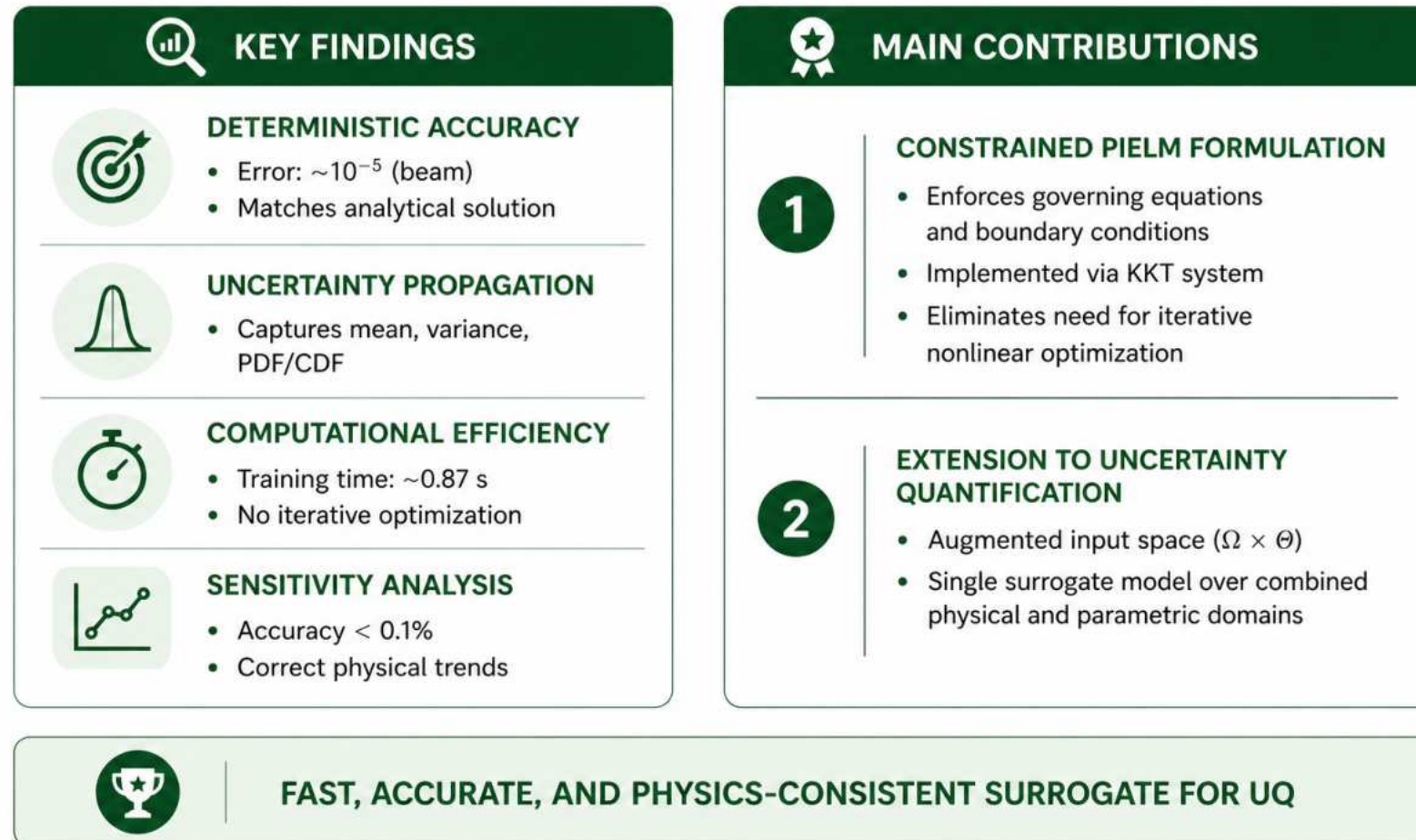
- Displacement is negatively sensitive to wave speed.
- Increase in c leads to a decrease in displacement.
- Sensitivity magnitude indicates moderate influence of **wave speed** on response.
- PIELM closely matches analytical sensitivity.



PIELM accurately captures both solution and parameter sensitivity.

Conclusions

KEY FINDINGS & MAIN CONTRIBUTIONS



Limitations

1 HIGH-DIMENSIONAL PROBLEMS



Number of required points grows very fast



Leads to high computational cost



Hard to scale to large dimensions

2 LIMITED TO LINEAR PDES



Works well for linear equations



Nonlinear PDEs require iterative extensions



Increases complexity and reduces efficiency

3 HYPERPARAMETER SENSITIVITY

Performance depends on careful selection of:



Number of neurons ($N_{neurons}$)



Number of collocation points ($N_{collocation}$)



Regularization parameter (λ)

Requires problem-dependent tuning

Future Work

These directions can extend the capabilities of the PIELM framework.



- **Adaptive Sampling:** Improve accuracy with fewer collocation points.



- **Nonlinear PDEs:** Extend to nonlinear problems using iterative methods.



- **Higher-Dimensional Uncertainty:** Use dimension reduction and sparse methods for high-dimensional spaces.



- **Enhanced Architectures:** Explore multi-layer or hybrid models to increase representation power.



- **Data Integration:** Combine physics-informed models with experimental data for better calibration and UQ.

- [1] Huang, G. B., Zhu, Q. Y., & Siew, C. K. (2006). Extreme learning machine: Theory and applications. *Neurocomputing*, 70(1-3), 489–501.
- [2] Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). Physics-informed neural networks. *Journal of Computational Physics*, 378, 686–707.
- [3] Dwivedi, V., & Srinivasan, B. (2020). Physics informed extreme learning machine (PIELM). *Neurocomputing*, 391, 96–118.
- [4] Xiu, D., & Karniadakis, G. E. (2002). The Wiener–Askey polynomial chaos for stochastic differential equations. *SIAM Journal on Scientific Computing*, 24(2), 619–644.
- [5] Williams, C. K., & Rasmussen, C. E. (2006). *Gaussian Processes for Machine Learning*. MIT Press.

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